Mongol Empire economic systems: Transmission without transformation

The Mongol Empire's economic achievement was unprecedented geographic integration without institutional depth—the largest land empire in history facilitated trade across Eurasia yet created no enduring financial innovations. This analysis reveals a critical paradox: **Peak-phase economic performance scored 3.5** (modest institutional development with significant facilitation effects), yet this "success" contained the seeds of collapse because Mongol fiscal systems were fundamentally extractive rather than developmental. The absence of financial innovation—no new instruments, no secondary markets, no central banking—confirms the CAS framework hypothesis that Economic/Fiscal/Financial domain development requires prior Foundational and Capability tier maturation that the Mongols never achieved.

The central finding challenges both traditional narratives: Mongol trade facilitation was neither purely incidental (passive barrier removal) nor fully institutional (active market-making comparable to Dutch VOC). Instead, the Mongols created a **hybrid system**—deliberately building infrastructure (yam postal network, standardized weights, paiza passports) while remaining dependent on existing merchant communities and financial instruments they transmitted rather than invented. This transmission-without-transformation pattern explains why there is no "Mongol moment" in financial history despite their unprecedented commercial reach.

Revenue extraction evolved from plunder to institutionalized predation

Rise phase (1206-1260): Conquest economics and emerging systematization

The early Mongol revenue system began as wartime plunder with gradual institutionalization. Under Genghis Khan, extraction was ad hoc—armies lived off conquered territories through direct seizure. The turning point came under **Ögödei Khan (1229-1241)**, who implemented the first systematic reforms. His Khitan advisor Yeh-lü Ch'u-ts'ai established fixed taxation replacing arbitrary plunder, yielding documented annual revenues of **500,000 liang of silver, 80,000 bolts of silk, and 400,000 tan of grain by 1231** (Grokipedia) (HIGH confidence—documented in Yuanshi).

The comprehensive empire-wide census of 1234-1236 assessed taxable populations across conquered territories, enabling standardized extraction. (IU Blogs) This census recorded household counts, number of men aged 15-60, fields, livestock, vineyards, and orchards— (Lumen Learning) creating the administrative infrastructure for systematic taxation.

Möngke Khan (1251-1259) attempted further centralization through extensive censuses across Iran, Afghanistan, Georgia, Armenia, Russia, Central Asia, and North China. He imposed fixed poll taxes collected by imperial agents, taxing wealthy families most severely. However, this provoked "popular riots and resistance in the western districts," revealing the tension between extraction intensity and political sustainability that would recur throughout Mongol rule.

The darughachi system: Structure and limitations

The **darughachi** (Mongolian) or **basqaq** (Turkic) served as Mongol overseers "in charge of taxes and administration in a certain province." Established under Genghis Khan from 1211, these officials performed tax collection, census administration, enforcement of Mongol law (Yasa), military logistics coordination, and prevention of local corruption.

Implementation varied significantly by region:

In China, darughachi implemented Yeh-lü Ch'u-ts'ai's reforms, achieving the revenue figures noted above. Under Yuan, the title "Zhangguan" replaced the darughachi designation, reflecting administrative evolution.

In Russian principalities, baskaki appeared in the 13th century "but were withdrawn by 1328 and the Grand Prince of Vladimir became the khan's tax collector." This delegation pattern proved crucial—Moscow's adoption of Mongol tax systems, with grand princes replacing basqaq with danshchiki officials, created the fiscal foundations for Muscovite expansion.

In Korea (Goryeo), 72 darughachi were reportedly sent after the 1231 invasion, but "repeated rebellions...made the stationing of darughachi difficult." The original darughachi were "all killed by Goryeo forces in the summer of 1232," demonstrating the system's vulnerability to coordinated resistance.

The darughachi system enabled "centralized fiscal oversight across conquered territories, standardizing tax collection through population censuses." However, insufficient personnel required delegation to local intermediaries, distance from central authority enabled corruption, and resistance in conquered territories limited effectiveness.

Ortaq partnerships: The Mongols' most sophisticated financial arrangement

The **ortaq** (Turkic: "partner") system represented the Mongol Empire's closest approximation to banking innovation. Elizabeth Endicott-West's foundational 1989 study "Merchant Associations in Yuan China: The Ortoγ" documented the structure:

Mongol investors (nobles, state treasury) provided capital in forms including metal coins, paper money, gold/silver ingots, and tradable goods. Detailed Pedia Muslim and Uyghur merchants conducted trade and money-lending operations. Wikipedia Malevus Profits were shared, with merchants receiving "very high commissions." The arrangement exhibited sophisticated features: liability concepts where principals were liable only to investment losses while ortoqs were liable for loan capital; risk distribution across multiple investors reducing individual exposure.

Government involvement intensified over time. In 1268, Kublai Khan created the **General Administration for the Supervision of Ortogh** to provide low-interest loans to merchant partners, marking peak state involvement. Merchants received special privileges including use of official relay stations (yam network) and the right to bear arms. Detailed Pedia

Enerelt Enkhbold's 2019 research documented a critical evolution: "As the Mongol rulers and nobles suffered more losses over time, loan capital became dominant over the capital structure of Mongol-ortoq partnerships."

This shifted business risks from principals to merchants, with ortoqs becoming "unlimitedly liable to loan capital." NomadIT

Was ortaq innovative? Critically, the ortaq system was **not** an innovation. Enkhbold's research definitively demonstrates that "the contractual arrangements of Mongol–ortoq partnerships closely resembled medieval partnership contracts, such as mudharaba, inan, societas and commenda." Taylor & Francis Online Taylor & Francis Online The Mongols adopted and developed existing Islamic and Mediterranean commercial forms, incorporating them into their unique multi-ethnic framework.

Peak phase revenue: Yuan fiscal architecture (1260-1294)

Under Kublai Khan, revenue systems achieved maximum sophistication. Key elements included:

Paper money (chao): The first unified paper currency backed by silver (1260-1276) enabled monetized taxation. Taxes became payable only in chao, forcing economic integration with the state monetary system.

Salt monopoly: Established in the 1270s, this became the primary revenue source. Herbert Franke's research shows salt tax sometimes exceeded **60-80% of total revenue**—a "strikingly modern" fiscal structure where agricultural tax contributed only ~30% of total revenue.

Finance Minister Ahmad Fanakati's policies (1262-1282) implemented "rigorous collection from non-Mongol subjects, including Han Chinese and Persian communities" while Mongol elites benefited from exemptions. Morris Rossabi provides "both the Confucian and the Muslim perspective" on Ahmad—efficient administrator to some, predatory extractor to others.

Regional variation in taxation

The Mongol Empire exhibited significant regional variation in tax structures:

Yuan Dynasty (China): Northern regions paid poll tax (dingfu) combined with triple tax in grain, material, and corvée labor. Southern regions (former Song territories) used the twice-taxation system (liangshuifa) with real-estate and profit taxes. Commercial taxes ran **3% for permanent merchants, 2% for itinerants; 10% trade duty pre-sale**.

Il-Khanate (Persia): Multiple taxes operated simultaneously—kharaj (land tax), qubchur (poll and herd tax), tamgha (commercial duties/sales tax), and taghar (provisions levies). Tax farming (moqāṭaʿa) became dominant, leading to "extortion and oppression." Encyclopaedia Iranica Gaykhatu Khan (1291-1295) "practically emptied the royal treasury with profligate spending."

Golden Horde (Russia): The dan (tribute) was the primary extraction mechanism from Russian principalities. Annual tribute estimates reached approximately **12-14 thousand rubles (roughly 1.5 tons of silver)**. Regional breakdown (mid-14th century): Vladimir 5,000 rubles; Suzdal-Nizhny Novgorod 1,500 rubles; Novgorod and Tver 2,000 rubles each; Moscow 1,280 rubles (MODERATE confidence—derived from chronicle sources).

Comparison to Ottoman timar: Why incentives matter

The comparison to the Ottoman timar system illuminates why Mongol extraction remained predatory rather than developmental.

The Ottoman timar was "a form of land grant or revenue assignment...conferred by the sultan to cavalrymen known as sipahis in exchange for military service and tax collection duties." Key features created incentive alignment:

- **Non-hereditary**: Grants were revocable, tied to active military service
- **Direct service-revenue link**: Sipahis kept set amounts as salary, delivered remainder to central state
- **Productivity incentives**: Timar holders benefited from "mak[ing] the lands they controlled more agriculturally productive" since "the more crops produced, the more taxes they could collect"
- **Cadastral surveys (tahrir)**: Periodic assessments ensured allocations aligned with yields

The Mongol system lacked these incentive-alignment mechanisms. Tax farmers, ortaq merchants, and darughachi had stronger incentives for short-term extraction than sustainable development. The compound interest rates on ortaq loans—reportedly reaching **100% annually** (Richmond Fed analysis)—exemplify this predatory orientation.

Assessment (HIGH confidence): Mongol extraction was primarily predatory with developmental elements varying by region and period. Unlike Ottoman timar which aligned incentives for land productivity, Mongol tax farming led to short-term extraction without long-term investment.

Yuan paper money: The world's first precious metal-backed fiat experiment

Introduction and three monetary phases

The Yuan dynasty represents a remarkable monetary experiment—the world's first precious metal-backed paper money system deployed as sole legal tender. The comprehensive 2024 study by Guan, Palma, and Wu in the *Economic History Review* provides the most rigorous quantitative analysis available. manchester

Paper money was introduced under Kublai Khan in 1260. In July 1260, he issued Jiaochao notes backed by silk (unsuccessful). In October 1260, he issued **zhongtongchao** (中統鈔), pegged to silver at a fixed rate of **2 guan zhongtongchao = 1 liang silver (~37g)**. This was the first precious metal standard backing paper money in world history.

The Yuan monetary system evolved through **three distinct phases**:

Phase 1: Full Silver Convertibility (1260-1275) Paper money was fully backed by silver reserves at fixed exchange rates. The government maintained silver reserves and bought back excess paper money with silver. 65 silver exchange bureaus were established (mostly in Eastern China), charging 3% commission. Paper money was exchangeable for silver at 70-80% of face value. This phase represented genuine monetary innovation in implementation—applying existing Song dynasty paper money concepts at unprecedented empire-wide scale with precious metal backing.

Phase 2: Nominal Silver Convertibility (1276-1309) Silver convertibility became nominal after 1276 when Southern Song conquest required massive issuance. The government integrated Southern Song huizi at 1:50 exchange rate, requiring enormous new zhongtongchao issuance. Silver reserves were transported to summer capital Khanbaliq, leaving local exchange bureaus often without silver available. In 1287, the first zhiyuanchao was issued, representing deliberate 80% devaluation of zhongtongchao.

Phase 3: Fiat Standard (1310-1368) From 1310 (zhidachao issuance), silver no longer played reserve role. The system became de jure fiat money. By 1352, the zhizhengchao was pure fiat currency, widely rejected by the population.

Circulation scale and inflation data

Quantitative estimates from Guan et al. (2024):

Metric	Value	Period
Circulation at Yuan proclamation	<2 million ding	1271
Circulation at collapse	>170 million ding	1368
Growth factor	~85-fold	Full dynasty
Peak annual issuance (zhidachao)	36 million ding	1310
Peak annual issuance (zhizhengchao)	50 million ding	1355

Inflation rates (based on respectability CPI constructed by Guan et al.):

- 1260-1290: **4.5% annual compound** (high early inflation)
- 1290-1340: **1.8% annual compound** (moderate, relatively stable)
- 1340-1355: **11% annual compound** (approaching hyperinflation)
- 1346-1355: **12.7% annual compound** (approximate doubling of price level) manchester

Key finding: The scholarly consensus now disputes characterization as true "hyperinflation" (defined as 50%+ monthly). Average inflation was approximately **5.2% per annum** by modern standards—high but not hyperinflationary (MODERATE confidence—dependent on price index construction methodology).

Why Yuan paper money collapsed

Primary cause: Civil warfare financing. Econometric analysis by Guan et al. shows that 1% increase in civil warfare correlated with 0.30% increase in nominal money issues. Under fiat standard: 1% increase in total warfare correlated with 0.61% increase in nominal money issues.

Challenging traditional narratives: The 2024 study found that imperial grants and natural disasters—traditionally blamed for over-issuance—were statistically insignificant drivers. The primary cause was military expenditure, particularly civil war financing. manchester

Fiscal dominance over monetary policy: Tax surplus data shows government consistently ran deficits financed by money printing:

- 1292: Tax surplus -659,000 ding; money issued 2,500,000 ding
- 1308: Tax surplus -9,400,000 ding; money issued 5,000,000 ding
- 1311: Tax surplus -14,000,000 ding; money issued 10,900,000 ding manchester

Structural failures:

1. Unlike later European systems (Bank of England after 1694), Yuan China lacked strong fiscal foundations and institutional credibility required for sustainable fiat money
2. Government could not effectively manage fiat circulation across vast territory manchester
3. Nine emperors in 1294-1333 period (average 4-year reign); ad hoc grants to secure noble support
4. Geographic limitations: Only 65 exchange bureaus; remote regions had minimal access to silver redemption

By 1356, paper money was rejected by the population. Markets reverted to barter economy. Contemporary scholar Wang Yun described the currency as "nothing but empty script."

Cross-khanate currency: No unified monetary space

Did a unified Mongol monetary space ever exist? Answer: **No** (HIGH confidence).

Il-Khanate (Persia): Gaykhatu Khan introduced chao modeled exactly on Yuan notes in 1294, even with Chinese characters. Yuan minister Bolad was sent to explain the system. Result: Complete failure within weeks/months due to public distrust of exotic currency, no tradition of paper money in Islamic world, and coercive implementation. The experiment caused "complete economic confusion." Ghazan's subsequent reforms (1295-1304) abandoned paper money and introduced unified bimetallic currency (gold dinars, silver dirhams).

Golden Horde: Maintained traditional metallic coinage following Islamic patterns. Lumen Learning Toqta Khan's 1310-11 reform unified silver coinage at standardized weight (~1.48-1.54g). No paper money ever adopted.

Chagatai Khanate: Kebek Khan's reforms (1318-1326) standardized coinage backed by silver reserves. Minted silver "kebeks" (~8g = 6 dirhams). No paper money adopted.

What DID exist: A common silver-based unit of account (sukhe/silver ingot) for long-distance trade. Akinobu Kuroda (2009) argues the Mongol regime established a "Eurasian Silver Century" (1276-1359) with common silver-based unit of account for long-distance exchanges. Cambridge Core However, this was monetary integration for trade settlement, not unified currency.

Pax Mongolica trade: Institutional facilitation without monopolistic extraction

Specific policies enabling trade

The Mongol Empire implemented deliberate trade facilitation through concrete institutional mechanisms:

Yam (Örtöö) Postal System: Established by Ögödei Khan in 1234 and expanded by successors. By late 13th century: approximately **10,000 stations with 300,000 horses** across Mongol-controlled territory (MODERATE confidence—derived from contemporary accounts). Stations positioned every 25-30 miles provided fresh horses, fodder, accommodations, and escorts. I Take History Messages could travel hundreds of miles per day—I Take History a letter from Beijing could reach Tabriz (5,000 miles) in approximately one month.

The Yassa (Great Law) codified strict commercial protections: theft outlawed with harsh penalties (restitution of 9x original value); empire-wide lost-and-found system established; religious freedom guaranteed (facilitating cross-cultural commerce).

Standardization measures: Unified system of weights and measures; standard trade tariffs; reduced border taxes; elimination of complex local taxation systems. Askpro

The paiza/gerege system

The paiza (from Chinese *paizi* 牌子; Mongolian: *gerege*) functioned as the world's first institutionalized passport system.

Hierarchy: Gold paizas for highest rank (special purposes); silver for mid-level officials/commanders; bronze/iron for governors/administrators; wood/basic for merchants, messengers, foreign ambassadors.

Privileges conferred: Right to requisition resources (room, board, fresh horses, escorts); access to yam relay stations; tax exemptions; safe passage throughout Mongol territories; ability to demand goods from civilian populations; protection under khan's authority.

Marco Polo received a foot-long, three-inch-wide golden tablet from Kublai Khan enabling his safe return to Venice. Federal Reserve Bank of Richm… Contemporary saying: "A maiden bearing a nugget of gold on her head could wander safely throughout the realm." Wikipedia

Quantitative trade evidence

Direct quantitative evidence remains limited but suggestive:

- Caravan records: 450 Muslim traders with 500 camels documented in single trading expedition
- Silver flows: London mint output increased from 7,000 kg (1278) to 60,000 kg (1280)—attributed partly to Mongol-facilitated trade
- Widespread distribution of Chinese porcelain reaching Iberian Peninsula

Proxy indicators: Cities participating in 13th-century world trade grew rapidly; proliferation of paper money circulation; development of bills of exchange, deposit banking, and insurance in Europe during this period.

Pre-Mongol vs. Mongol comparison: Historian Warwick Ball argues there was "no coherent overland trade system and no free movement of goods from East Asia to the West until the period of the Mongol Empire." Wikipedia First documented direct personal contact between Europe and China occurred in the 13th century (Morris Rossabi).

Maritime trade under Yuan

Angela Schottenhammer's research documents active Yuan sponsorship of maritime commerce:

Quanzhou rose to prominence as international trading port, called the "Emporium of the World." Archaeological evidence includes Song Dynasty shipwreck, mosques, Hindu sites, Tamil merchant guild inscriptions.

Yuan maritime policy: Government actively promoted maritime trade; strong interest in strengthening China's role in Asian maritime networks; diplomatic missions accompanied search for trade relations; Mongols expanded Grand Canal from southern China to Daidu (Beijing). China Knowledge

Kublai Khan's initiatives included massive expansion of China's maritime fleet and extension of Maritime Silk Road to Europe by sea (previously only overland to Istanbul).

Testing Hypothesis 1: Incidental vs. institutional facilitation

Evidence for INCIDENTAL (passive barrier removal):

- Mongols did not fundamentally transform production systems
- Relied on existing merchants (Muslims, Uyghurs) rather than creating commercial enterprises
- Did not establish factories, plantations, or production facilities
- Security benefits were byproduct of military control

Evidence for INSTITUTIONAL (active creation):

- Deliberate construction of yam system
- Systematic issuance of paizas
- Capital investment through ortaq partnerships
- Standardization policies across empire
- Active recruitment and elevation of merchant class
- Rossabi: Kublai Khan "promoted the construction of roads and expanded the postal-station system to assist in the development of commerce" Encyclopedia.com

Comparison to Dutch VOC: The VOC (1602-1799) employed 57,000 people at peak with 150 merchant ships and 40 warships. It created monopolies through military force, employed "trade-by-arms" policy, replaced local trading networks with Dutch-controlled systems, and manipulated markets by controlling supply. The VOC engaged in colonial violence with "upwards of one million Indonesian people" killed for economic/political gain.

Assessment: Mongol trade facilitation was **INSTITUTIONAL but non-extractive**—actively creating infrastructure and standardization without the monopolistic extraction characteristic of European trading companies. The Mongols invested in trade networks but did not monopolize production or eliminate competitors. They were "active participants in Eurasian trade, both as investors and consumers" but operated through partnership rather than monopoly.

Hypothesis 1 verdict (MODERATE confidence): PARTIALLY REJECTED. Evidence supports institutional activity, though the *nature* of institutions (decentralized, partnership-based) differs from Dutch VOC-style centralized market-making. The Mongols created a different KIND of institutional framework, not an absence of one.

Financial innovation: Transmission without transformation

Did Mongols create ANY new financial instruments?

This section addresses the critical question of Mongol financial innovation by systematic comparison to European developments.

Venetian innovations (12th-14th centuries)

Venice pioneered fundamental financial innovations during the same period as Mongol dominance:

Prestiti (Government Bonds, 1262): Consolidated state debt into tradeable perpetual obligations; fixed 5% interest paid semiannually for over 100 years; created secondary market for sovereign bonds; formed Monte Vecchio fund structure.

Colleganza partnerships: Limited-liability partnerships for maritime trade; one party provided capital, another provided labor; "viewed by all economic historians as one of the key commercial innovations of medieval times" (Lopez and Raymond, 1967); enabled pooling of capital from multiple investors for risky ventures.

Institutional features absent in Mongol system: Tradeable securities with organized secondary markets; parliamentary oversight of public finance; contractual enforcement through specialized commercial courts; standardized accounting practices (double-entry bookkeeping developed in Italy).

Dutch innovations (17th century)

VOC (1602): First joint-stock company with permanent capital; first publicly traded shares on Amsterdam Stock Exchange; limited liability for all shareholders; tradeable shares enabling price discovery.

Amsterdam Wisselbank (1609): Centralized exchange bank; deposit banking with standardized currency.

Derivatives trading: Options and futures on VOC shares by mid-17th century.

British Financial Revolution (1688-1720)

Bank of England (1694): Central bank managing national debt; parliamentary control of public finance; long-term government bonds with credible repayment. North and Weingast (1989) argue the key innovation was removal of "time-inconsistent policy making" by "divesting public finance from the Crown's control" through parliamentary oversight.

What Europeans created that Mongols did not

Innovation	European Origin	Mongol Equivalent?
Tradeable government bonds	Venice 1262	None
Limited liability corporations	Venice/Netherlands	Ortaq similar but not transferable
Organized stock exchange	Amsterdam 1602	None
Central banking	England 1694	None
Options/futures trading	Amsterdam 17th c.	None
Perpetual annuities	Venice/England	None
Parliamentary fiscal control	England 1693	None

Testing Hypothesis 2: No financial innovation

Evidence SUPPORTING the hypothesis (Mongols as transmitters):

1. **Ortaq = Adopted structure**: Enkhbold (2019) definitively demonstrates ortaq "closely resembled medieval partnership contracts, such as mudharaba, inan, societas and commenda" Taylor & Francis Online
2. **Paper money = Chinese invention**: Song dynasty jiaozi preceded Yuan chao by centuries
3. **Hawala/Mudaraba = Pre-existing Islamic instruments**: Documented in Islamic jurisprudence since 8th century
4. **No secondary markets**: Unlike Venetian prestiti, Mongol financial instruments were not tradeable
5. **No institutional permanence**: Financial arrangements dissolved with the empire

Possible counter-evidence:

1. **Scale innovation**: First empire-wide unified currency system
2. **Silver standard for paper money**: Yuan first to back paper currency with precious metals (though adaptation, not conceptual innovation)
3. **Multi-ethnic financial integration**: Unprecedented combination of Chinese, Persian, and Turkic financial practices

Hypothesis 2 verdict (HIGH confidence): LARGELY SUPPORTED. Mongols primarily transmitted and scaled existing instruments rather than creating wholly new forms. The **scale of application** and **institutional integration** represent meaningful innovation in implementation, but no genuinely new financial forms were created.

Why no "Mongol moment" in financial history?

Nomadic mobility vs. fixed institutions: The Mongols' pastoral lifestyle was structurally incompatible with institution-building. Their view of empire as "the heirloom of the imperial clan as a whole" rather than permanent state apparatus precluded durable institutional development. Encyclopedia Britannica

Absence of legal infrastructure: European financial innovation required specialized commercial courts (absent in Mongol system), enforceable contract law across jurisdictions, and property rights protection. Mongol appanage system was fluid and contestable. Encyclopedia Britannica

Political economy structure: North and Weingast's framework explains British success through credible commitment via parliamentary control. Mongol political authority remained personalized around khans, subject to kurultai succession disputes, and fragmented across successor khanates after 1260.

Short time horizon: The unified empire lasted roughly 1206-1260 (54 years) before fragmenting. The Pax Mongolica period (c.1250-1350) was too brief for institutional maturation. By contrast, Venice developed financial institutions over 400+ years.

Janet Abu-Lughod's world system thesis: Evidence and critiques

Core argument

In *Before European Hegemony: The World System A.D. 1250-1350* (1989), Abu-Lughod argued that a pre-modern world system existed in the 13th century, predating Wallerstein's 16th-century Eurocentric world-system by 300 years. This system was **polycentric** (no single hegemon) with three core regions: China (most advanced), Arab World (Egypt/Near East), and Western Europe.

Eight interlocking circuits of trade connected northwest Europe to China. Critical argument: "There was no inherent historical necessity that shifted the system to favor the West rather than the East." The West "pulled ahead because the 'Orient' was temporarily in disarray."

Supporting evidence

- Documented trade networks linking eight interlocking subsystems
- City-based analysis showing commercial interconnection
- Use of navigational manuals, warehouse receipts, mortality figures, notarized contracts
- Demonstrated "interdependent price movements" across regions
- Won American Sociological Association Award (1990) for best book in Political Economy

Scholarly critiques

Methodological weaknesses:

- "Lacks effective argumentation and deep economic analysis"
- Relies heavily on political history rather than original economic analysis
- "Thesis is not backed by empirical evidence" regarding quantitative trade volumes

Conceptual issues:

- Whether genuine integrated "world system" existed or merely connected trade networks
- Parameters for comparing European "lag" not clearly explained
- Decline causation remains underdeveloped
- Does not convincingly explain why the 13th-century system collapsed

Alternative positions:

- Chase-Dunn and Hall (1997): Argued for "many world-systems of various kinds"

- Andre Gunder Frank: Disagreed on periodization—argued one system existed for 5,000 years
- Wallerstein maintained 16th-century origins position

Assessment (MODERATE confidence): Abu-Lughod's thesis remains influential but contested. The evidence supports increased commercial integration during the Mongol period, but whether this constituted a genuine "world system" with structural interdependence (rather than connected trade networks) remains debated.

Testing Hypothesis 3: Extractive vs. developmental extraction

Evidence for EXTRACTIVE character

1. **High interest rates**: Ortaq lending at "compound annual rate of 100 percent" (Richmond Fed)
2. **Appanage abuses**: Holders "demanded excessive revenues and freed themselves from taxes"
3. **Heavy indirect taxation**: Salt tax sometimes 60-80% of total revenue
4. **Fiscal-driven monetary policy**: Paper money over-issuance driven by deficits, not stability
5. **Military orientation**: "Military character of Mongol rule" prioritized extraction for war
6. **Conquest devastation**: "Large parts of the rural population were killed or enslaved"
7. **Elite capture**: "The profit from the trade along the Inner Asian caravan routes went into the hands of the Mongol ruling elite and did not contribute to ensure the wealth of the common Chinese people"

Evidence for DEVELOPMENTAL elements

1. **Agricultural policy**: Rossabi documents Khubilai "actively seeking to build the economies of the conquered peoples"
2. **Office for Stimulation of Agriculture (1261)**: Eight officials appointed for agricultural improvement
3. **Land reclamation programs**: Yuan policy encouraged bringing wasteland under cultivation
4. **Tax remissions**: Documented during natural disasters
5. **Trade infrastructure**: Roads, postal stations had developmental effects
6. **Merchant elevation**: "The status of merchants rose during that period and China never went back" (Rossabi)
7. **Legal moderation**: Yuan legal code had "half the number of capital crimes as the Song dynasty code"

Comparative assessment: Mongol vs. Ottoman incentive structures

Feature	Mongol System	Ottoman Timar
Land ownership	State/appanage holders	State (miri lands)
Revenue assignment	Tax farming, direct extraction	Revenue grants tied to service
Military service link	Loose; separate darughachi/tammachi	Direct; sipahi military obligation
Heritability	Appanages sometimes inherited	Non-hereditary; performance-based
Long-term investment incentive	Low (extraction-focused)	Higher (productivity benefits holder)
Oversight mechanism	Darughachi supervision, often distant	Kadı courts, tahrir surveys

Hypothesis 3 verdict (HIGH confidence): LARGELY SUPPORTED. While developmental policies existed (especially under Khubilai), the overall system remained primarily extractive. Unlike Ottoman timar which aligned incentives toward land productivity, Mongol tax farming through ortoq partnerships prioritized short-term returns. The 100% compound interest rates and salt tax dominance suggest predatory rather than growth-oriented fiscal policy.

Scholarly engagement: Key interpretations synthesized

Thomas T. Allsen (1940-2019)

Allsen's *Commodity and Exchange in the Mongol Empire* (1997) and *Culture and Conquest in Mongol Eurasia* (2001) fundamentally reframed understanding of Mongol economic activity. He argued that "one people's luxury is another's necessity," challenging the traditional luxury vs. necessity trade distinction.

Key finding: The Mongols were **active stimulators** of production, not mere passive facilitators. "High levels of consumption of luxury textiles at the Mongol courts not only meant they took an active role in encouraging trade but also led them to stimulate production." He documented systematic "traffic in human talent"—Mongols "systematically identified and mobilized artisans of diverse backgrounds and frequently transported them from one cultural zone of the empire to another."

The empire functioned as a "cultural clearing house for the Old World"—deliberately sponsoring exchange of technologies, commodities, and expertise.

Elizabeth Endicott-West

Mongol Rule in China documented Yuan bureaucratic structures and the "basic dilemma of Mongol rule in China"—their "inability to achieve a durable identification with Chinese civilian institutions." Her 1989 *Asia*

Major article on ortaq partnerships remains the foundational English-language study of the system.

Morris Rossabi

Khubilai Khan: His Life and Times (1988) presents Khubilai as a pragmatic administrator balancing Mongol and Chinese traditions. Key economic arguments: agricultural focus on peasant welfare; trade infrastructure expansion; paper money as unprecedented monetary experiment; elevation of merchant social status.

David Morgan

The Mongols (1986, 2007) provides a "largely negative assessment of the Mongol impact" emphasizing destruction, particularly in Iran where Mongols destroyed "sophisticated underground network of irrigation channels." However, the 2007 second edition notably admits his assessment "looks very old-fashioned" and that "the subject now has a distinctly different feel to it"—acknowledging the scholarly reassessment led by Allsen.

Herbert Franke

Franke's contribution to *Cambridge History of China*, Volume 6 documented Yuan fiscal policy showing the "strikingly modern" fiscal structure where agricultural tax contributed only ~30% of total revenue, with salt monopoly and commercial taxes providing the majority.

Angela Schottenhammer

Her research on Yuan maritime trade, particularly *The Emporium of the World: Maritime Quanzhou, 1000-1400* (2001), demonstrates that Yuan "vividly sponsored" maritime commerce—challenging the incidental facilitation thesis for the maritime sphere.

Areas of scholarly consensus

1. **Increased trade volume**: All scholars agree Pax Mongolica saw unprecedented transcontinental trade increases
2. **Paper money significance**: Consensus that Yuan paper money was historically significant as first attempt at sole legal tender backed by precious metals
3. **Short-term success, long-term failure**: System worked well 1260-1294, deteriorated significantly afterward
4. **Cultural exchange**: Universal acknowledgment of unprecedented technology/knowledge transfer

Areas of scholarly disagreement

Issue	Position A	Position B
Overall impact	Morgan: "essentially destructive"	Allsen: "champions of cross-cultural contacts"

Issue	Position A	Position B
Institutional development	Passive facilitation	Active institutional creation
World system periodization	Abu-Lughod: 13th century	Wallerstein: 16th century
Fiscal sustainability	Extractive/predatory	Initially well-structured

Phase-by-phase scoring with detailed justification

Rise phase (1206-1260): Score 2.5/5.0

Justification:

Positive factors:

- Emerging systematization under Ögödei (census, standardized taxation)
- Initial ortaq partnerships facilitating trade
- Yam system construction began
- Revenue of 500,000 liang silver annually documented

Negative factors:

- Massive economic destruction across Central Asia, Iran, North China
- Trade disruption during conquest exceeded facilitation benefits
- Limited institutional development
- Extraction remained ad hoc in many regions

Data quality: MODERATE—conquest-era documentation fragmentary

Confidence: MODERATE

Peak phase (1260-1294): Score 3.5/5.0

Justification:

Positive factors:

- Paper money successfully stabilized (1260-1276) with genuine silver convertibility
- Agricultural development policies (Office for Stimulation of Agriculture)

- Maritime commerce expansion under active state sponsorship
- Ortaq system at full institutional operation
- Trade infrastructure (yam) functioning at maximum extent
- Abu-Lughod's "world system" at height
- Merchant status elevated above traditional Chinese hierarchy
- Multiple tax reforms attempting rationalization

Negative factors:

- Seeds of fiscal problems present from 1276 (nominal convertibility)
- Extractive orientation of ortaq partnerships (100% interest rates)
- Salt tax dominance suggesting regressive fiscal structure
- No genuine financial innovation—transmission of existing instruments
- Appanage system creating fragmented incentives
- High inflation (4.5% annually) even during "stable" period

Comparison to other empires: Score of 3.5 reflects significant trade facilitation and monetary experimentation, but falls short of true institutional development. Ottoman timar system would score higher on incentive alignment; Song dynasty scored higher on fiscal sophistication and commercial taxation.

Data quality: HIGH for Yuan China; MODERATE for other khanates

Confidence: HIGH

Decline phase (1294-1368): Score 2.0/5.0

Justification:

Negative factors:

- Fiscal collapse driving accelerating inflation (11% annually 1340-1355)
- Paper money transition to worthless fiat (rejected by 1356)
- Political fragmentation (nine emperors 1294-1333)
- Silver standard fully abandoned (1310)
- Civil war financing driving monetary over-issuance
- Black Death spread via trade routes (killing ~1/3 of China's population)
- Trade contraction and route disruption

- Inter-khanate conflicts (Golden Horde vs. Il-Khanate)

Mitigating factors:

- Trade continued at reduced levels
- Some regional prosperity maintained
- Inflation, while high, technically not hyperinflationary by modern standards

Data quality: MODERATE to HIGH—Yuanshi documentation incomplete after 1332

Confidence: HIGH

Domain trajectory summary

Phase	Score	Confidence	Key Characteristic
Rise (1206-1260)	2.5	MODERATE	Conquest economics with emerging systematization
Peak (1260-1294)	3.5	HIGH	Maximum facilitation without institutional depth
Decline (1294-1368)	2.0	HIGH	Fiscal collapse and monetary disintegration

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Quantitative data synthesis

Revenue data (MODERATE confidence)

Metric	Value	Period
Yuan annual revenue (Yeh-lü reforms)	500,000 liang silver + 80,000 bolts silk + 400,000 tan grain	1231
Golden Horde annual Russian tribute	12-14,000 rubles (~1.5 tons silver)	Mid-14th century
Salt tax share of Yuan revenue	60-80%	Peak periods
Agricultural tax share	~30%	1328 (only comprehensive year)

Monetary data (HIGH confidence for circulation; MODERATE for inflation)

Metric	Value	Period
Paper money circulation	<2 million ding	1271
Paper money circulation	>170 million ding	1368
Growth factor	~85-fold	1271-1368
Inflation (stable period)	1.8% annually	1290-1340
Inflation (collapse period)	11-12.7% annually	1340-1355

Trade infrastructure (MODERATE confidence)

Metric	Value
Yam stations	~10,000
Yam horses	~300,000
Station spacing	25-30 miles
Silver exchange bureaus	65

Commercial tax rates (HIGH confidence)

Tax	Rate
Permanent merchants	3%
Itinerant merchants	2%
Pre-sale trade duty	10%
Tamga (Golden Horde)	3-5%

Framework implications: Testing CAS culmination tier theory

The Mongol case provides a critical test for the CAS Multi-Domain Framework hypothesis that

Economic/Fiscal/Financial development represents a "culmination" requiring prior Foundational and Capability tier development.

Evidence supporting the framework

1. **Military excellence without fiscal sustainability**: Despite Military domain scoring 5.0 (highest ever), fiscal systems collapsed within decades of peak military achievement
2. **Educational absence correlated with institutional weakness**: Educational domain score of 1.5 (complete institutional absence) correlates with inability to develop enduring financial institutions
3. **Transmission without transformation**: Without educational/administrative foundations, Mongols could transmit existing financial instruments but not create new ones
4. **Short-term extraction vs. long-term development**: Absence of Foundational tier development (education, civil administration) led to extractive rather than developmental economic orientation

Framework anomalies

1. **Peak trade facilitation despite institutional thinness**: Peak phase score of 3.5 demonstrates that significant economic facilitation is possible without deep institutional development—challenging the framework's sequential requirement
2. **Hybrid institutional model**: Mongol reliance on existing merchant communities and transmitted instruments represents a "borrowing" strategy not fully captured by the framework
3. **Regional variation**: Different khanates achieved different institutional depths (Yuan more developed than Chagatai), suggesting domain scores should be disaggregated by political unit

Conclusion for framework

The Mongol case **largely supports** the CAS framework proposition that Economic/Fiscal/Financial domain development requires prior Foundational and Capability tier maturation. The Mongols achieved peak trade facilitation (3.5) but could not sustain it precisely because they lacked the educational and administrative foundations for genuine institutional development. Their financial "innovation" was transmission at scale, not creation. The collapse from 3.5 to 2.0 within 74 years demonstrates the instability of economic achievement without deeper institutional roots.

The central thesis being tested—"Can military excellence alone sustain hegemony?"—receives a definitive negative answer from the economic domain. Military conquest could create the conditions for trade facilitation, but without fiscal sustainability grounded in developmental (rather than extractive) institutions, the economic achievements proved ephemeral.

Data quality limitations and confidence summary

High confidence claims

- Three-phase monetary standard evolution (full silver → nominal → fiat)
- Ortaq system borrowed from Islamic/Mediterranean precedents (not innovation)
- No unified monetary space across khanates
- Paper money rejected by 1356, leading to barter economy
- European financial innovations (bonds, joint-stock companies) were genuinely novel forms absent from Mongol system

Moderate confidence claims

- Specific revenue figures (derived from Yuanshi with inherent biases)
- Inflation rate calculations (dependent on price index construction)
- Trade volume increases (based on proxy indicators, not direct measurement)
- Yam system extent (contemporary accounts may exaggerate)
- Abu-Lughod's "world system" characterization

Low confidence claims

- Empire-wide aggregate fiscal data
- Extraction rates as percentage of production
- Precise capital scales in ortaq partnerships
- Counterfactual trade volumes without Mongol conquest

Primary source limitations

Key sources include Secret History of the Mongols, Rashid al-Din's histories, Yuanshi, Marco Polo's accounts, and various Russian chronicles. Modern scholarship relies heavily on Thomas Allsen, Elizabeth Endicott-West, Herbert Franke, and the 2024 Guan-Palma-Wu econometric study. Economic data is especially scarce for Central Asian territories and the post-1332 period.

Conclusion: Facilitation without foundation

The Mongol Empire's economic legacy was **transmission without transformation**—unprecedented geographic integration of existing commercial practices without creation of enduring financial innovations. The Yuan paper money experiment, while historically significant as the first precious metal-backed sole legal tender, ultimately failed because fiscal discipline could not survive military pressures. The ortaq system, while sophisticated, borrowed from Islamic mudharaba and Mediterranean commenda without conceptual innovation.

The three hypotheses tested receive the following verdicts:

1. **Trade facilitation was incidental, not institutional**: PARTIALLY REJECTED (MODERATE confidence). Evidence supports institutional activity—yam system, paiza credentials, ortaq partnerships, standardization—though different in kind from Dutch VOC-style monopolistic market-making.
2. **No financial innovation occurred**: LARGELY SUPPORTED (HIGH confidence). Mongols transmitted and scaled existing Chinese and Islamic instruments but created no genuinely new financial forms. This explains the absence of a "Mongol moment" in financial history.
3. **Revenue extraction was extractive, not developmental**: LARGELY SUPPORTED (HIGH confidence). Despite developmental policies under Khubilai, overall system prioritized short-term extraction. Unlike Ottoman timar, Mongol fiscal arrangements did not align incentives for long-term productivity investment.

The Economic/Fiscal/Financial domain scores—2.5 (Rise), 3.5 (Peak), 2.0 (Decline)—reveal a pattern of rapid achievement and rapid collapse characteristic of extraction-based systems. The peak score of 3.5 represents genuine accomplishment in trade facilitation, but its unsustainability confirms the CAS framework insight that economic domain development requires deeper institutional foundations than the Mongols ever achieved.

The Mongol case thus demonstrates both the possibilities and limits of military conquest as economic driver. Conquest could remove barriers, create security, and facilitate exchange—but it could not create the legal infrastructure, property rights protection, and aligned incentive structures that characterize developmental (rather than extractive) economic institutions. The absence of these foundations explains why the largest land empire in history left no enduring mark on financial history.